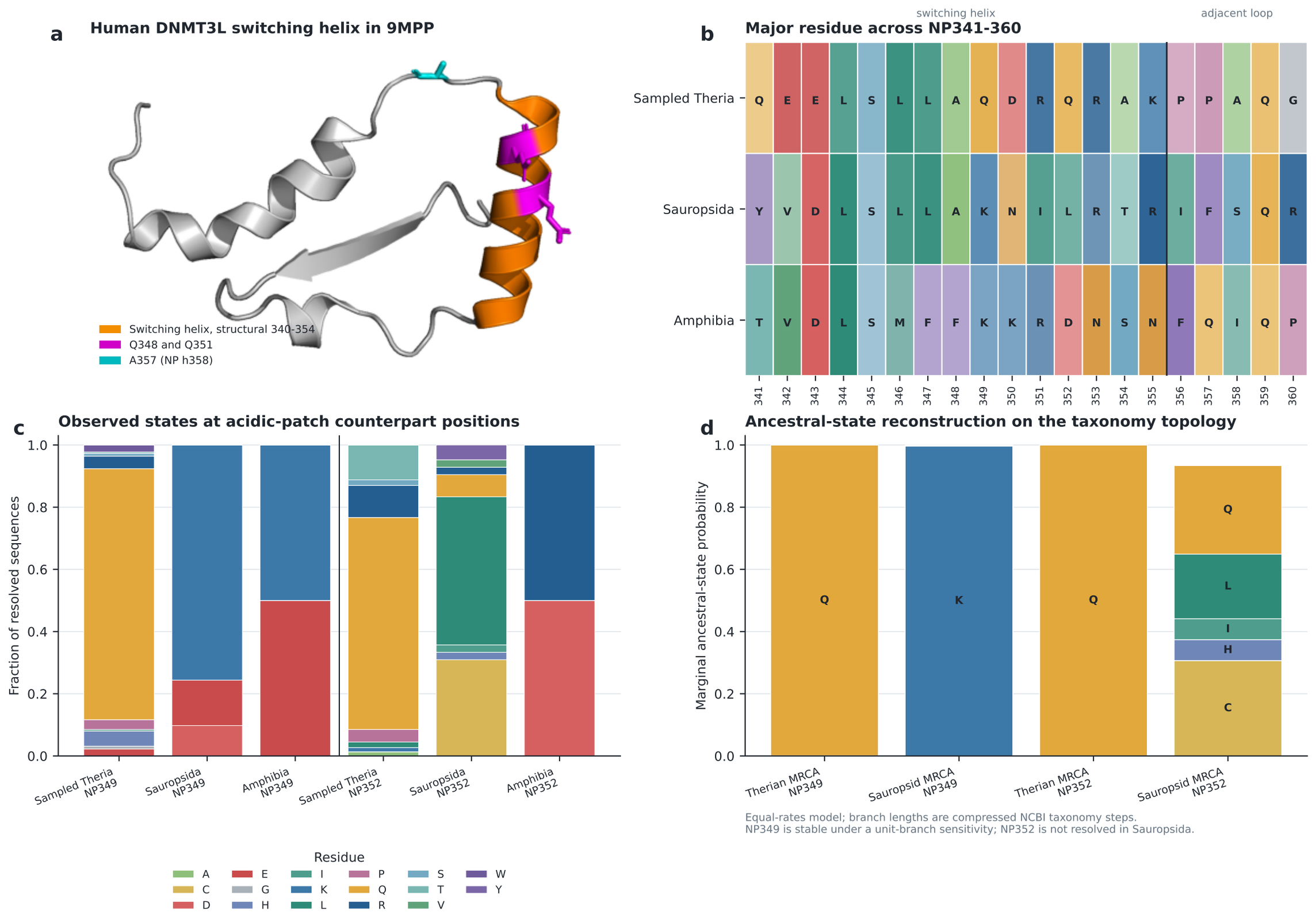

a
Human DNMT3L switching helix in 9MPP
Switching helix, structural 340-354
Q348 and Q351
A357 (NP h358)
b
Major residue across NP341-360
switching helix
adjacent loop
Sampled Theria
Sauropsida
Amphibia
Q E E L S L L A Q D R Q R A K P P A Q G
Y V D L S L L A K N I L R T R I F S Q R
T V D L S M F F K K R D N S N F Q I Q P
341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
c
Observed states at acidic-patch counterpart positions
Fraction of resolved sequences
0.0 0.2 0.4 0.6 0.8 1.0
Sampled Theria NP349
Sauropsida NP349
Amphibia NP349
Sampled Theria NP352
Sauropsida NP352
Amphibia NP352
Residue
A C D E G H I K L P Q R S T V W Y
d
Ancestral-state reconstruction on the taxonomy topology
Marginal ancestral-state probability
0.0 0.2 0.4 0.6 0.8 1.0
Q
K
Q
Q L I H C
Therian MRCA NP349
Sauropsid MRCA NP349
Therian MRCA NP352
Sauropsid MRCA NP352
Equal-rates model; branch lengths are compressed NCBI taxonomy steps.
NP349 is stable under a unit-branch sensitivity; NP352 is not resolved in Sauropsida.